

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination: 2022 - 23 Program: B. Tech. Branch: Mechanical Engineering Semester: III Course Code & Name: BTMEC302 Materials Science and Metallurgy Max. Marks: 60 Date: 15.3.2023 Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculator is permitted. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is atomic packing factor? Prove that the atomic packing factor for FCC structure is 0.74.	Understand	6
B)	What are the mechanisms of plastic deformation? Explain slip mechanism due to the movement of edge dislocations and screw dislocations with neat sketch.	Understand	6
C)	Discuss Vickers hardness test w.r.t working principle, indenter details, formula, advantages and limitations.	Remember	6
Q.2	Solve Any Two of the following.		12
A)	Draw Iron-Carbide Equilibrium diagram and define all the phases.	Remember	6
B)	Explain Hume-Rothery's rules of solid solubility. What is Gibb's phase rule.	Remember	6
C)	What is the importance of TTT diagram? Explain the procedure to determine these diagrams with the help of neat sketch.	Understand	6
Q. 3	Solve Any Two of the following.		12
A)	What are the objectives of heat treatment? Explain different types of annealing processes with the help of schematic diagrams.	Understand	6
B)	Define hardenability and explain, in detail, the Jominy End Quench Test with neat sketch.	Remember	6
C)	Explain induction hardening process in detail. Give its advantages and limitations.	Understand	6
Q.4	Solve Any Two of the following.		12
A)	Describe various steps in specimen preparation for microscopy.	Understand	6
B)	Explain the construction and working principle of metallurgical microscope with neat sketch.	Remember	6
C)	Discuss Spark test in detail and draw the spark pattern for the Mild Steel and High Carbon Steel.	Understand	6

Q. 5	Solve <i>Any Two</i> of the following.		12
A)	Describe Magnetic Particle testing w.r.t. principle of working and applications.	Understand	6
B)	Explain Dye Penetrant Test in detail w.r.t. basic principle, steps, and limitations.	Understand	6
C)	Explain basic mechanism of dispersion strengthening. What are its critical factors, advantages and applications?	Remember	6
*** End ***			

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